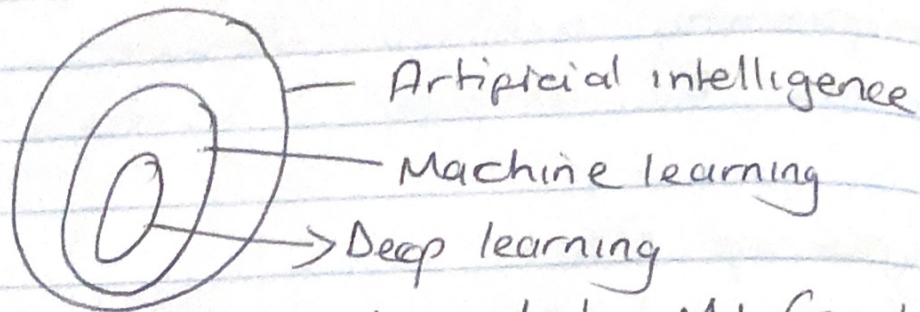


Question 13

- a) Deep learning is a subset of ML that is specialised to use multi-layered neural networks to automatically identify patterns that are complex in large amounts of data. It relates to Artificial Intelligence (AI) and Machine Learning (ML) in that it is a subset as shown in the diagram below:



AI is the broadest umbrella, followed by ML (a subfield) followed by Deep learning (a specialised subfield of ML).

b) Neural Networks

These are computing systems that drive deep learning. They are inspired by the biological network of human brains. They consist of numerous interconnected nodes that are organised in layers.

Neural Networks work through an iterative process of trial and error through forward propagation, calculating error, back propagation and optimisation.